

HDOC - Day 21

Nested-Loop Arithmetic Tables for a Range of Integers

Algorithm

1. Start. 2. Read positive integers m and n and validate $m \leq n$. 3. Display the five-option menu. 4. Option 1: for every integer from m to n, print its first 10 multiples using a nested loop. 5. Option 2: print its addition table using addends 1 to 10. 6. Option 3: print its subtraction table using subtrahends 1 to 10. 7. Option 4: divide each integer by 1 through 10 and print results to two decimal places. 8. For an invalid menu choice, display an error. 9. Repeat until option 5 (Exit). 10. Stop.

Test Case Table

TC	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	m=3,n=5; choice 1	10 multiples for 3,4,5	Matches expected	Pass
2	m=2,n=3; choice 2	Addition tables 1-10	Matches expected	Pass
3	m=2,n=3; choice 3	Subtraction tables 1-10	Matches expected	Pass
4	m=3,n=4; choice 4	Division results to 2 decimals	Matches expected	Pass
5	m=3,n=5; choice 9	Invalid menu choice	Invalid menu choice	Pass

C Program

```
#include <stdio.h>
int main() {
    int m,n,choice,i,j;
    do {
        printf("Enter m and n (m <= n): ");
        scanf("%d%d",&m,&n);
        if(m<=0 || n<=0 || m>n) printf("Invalid input. Enter positive m and n with m <= n.\n");
    } while(m<=0 || n<=0 || m>n);

    do {
        printf("\n1. First 10 multiples\n2. Addition tables\n3. Subtraction tables\n");
        printf("4. Division results\n5. Exit\nEnter choice: ");
        scanf("%d",&choice);

        switch(choice) {
            case 1:
                printf("First 10 multiples\n");
                for(i=m;i<=n;i++) {
                    printf("%d: ",i);
                    for(j=1;j<=10;j++) printf("%d ",i*j);
                    printf("\n");
                }
                break;

            case 2:
                for(i=m;i<=n;i++) {
                    printf("\nAddition table for %d:\n",i);
                    for(j=1;j<=10;j++) printf("%d + %d = %d\n",i,j,i+j);
                }
                break;

            case 3:
                for(i=m;i<=n;i++) {
                    printf("\nSubtraction table for %d:\n",i);
                    for(j=1;j<=10;j++) printf("%d - %d = %d\n",i,j,i-j);
                }
                break;

            case 4:
                for(i=m;i<=n;i++) {
                    printf("\nDivision results for %d:\n",i);
                    for(j=1;j<=10;j++) printf("%d / %d = %.2f\n",i,j,(float)i/j);
                }
                break;

            case 5:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid menu choice.\n");
        }
    } while(choice!=5);
    return 0;
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day21.c -o day21
puppie@Anvikas-MacBook-Air Desktop % ./day21
Enter m and n (m <= n): 3 5
Enter choice: 1
First 10 multiples
3: 3 6 9 12 15 18 21 24 27 30
4: 4 8 12 16 20 24 28 32 36 40
5: 5 10 15 20 25 30 35 40 45 50
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day21
Enter m and n (m <= n): 2 3
Enter choice: 2
Addition table for 2:
2 + 1 = 3 ... 2 + 10 = 12
Addition table for 3:
3 + 1 = 4 ... 3 + 10 = 13
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day21
Enter m and n (m <= n): 2 3
Enter choice: 3
Subtraction table for 2:
2 - 1 = 1 ... 2 - 10 = -8
Subtraction table for 3:
3 - 1 = 2 ... 3 - 10 = -7
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day21
Enter m and n (m <= n): 3 4
Enter choice: 4
Division results for 3:
3 / 1 = 3.00 ... 3 / 10 = 0.30
Division results for 4:
4 / 1 = 4.00 ... 4 / 10 = 0.40
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day21
Enter m and n (m <= n): 3 5
Enter choice: 9
Invalid menu choice.
```

Result: The menu-driven C program was compiled, executed and tested successfully for multiples, addition, subtraction, division, and invalid menu input.